> sentence. Much as we might wish to believe otherwise, universal love and the welfare of the species as a whole are concepts which simply do not make evolutionary sense.

We may deplore these truths, Dawkins says, but that does not make them any less true. The more clearly we understand the selfishness of the genetic process, however, the better qualified we shall be to *teach* the merits of generosity and co-operativeness and all else that works for the common good, and Dawkins expounds more clearly than most the special importance in mankind of cultural or 'exogenetic' evolution.

In his last and most important chapter, Dawkins challenges himself to formulate one fundamental principle that would certainly apply to all evolutionary systems—even perhaps to organisms in which silicon atoms took the place of carbon atoms, and to organisms like human beings in which so much of evolution is mediated through non-genetic channels. The principle is that of evolution through the net reproductive advantage of replicating entities. For ordinary organisms under ordinary circumstances these entities are the singularities in DNA molecules known as 'genes.' For Dawkins the unit of cultural transmission is that which he calls the 'meme' and in his last chapter he expounds what is in effect a Darwinian theory of memes.

To Dawkins's exhilaratingly good book I will add one footnote: the idea that the possession of a memory function is a fundamental attribute of all living things was first propounded by an Austrian physiologist Ewald Hering in 1870. He spoke of his unit as the 'mneme,' a word of conscious etymological rectitude. Richard Semon's exposition of the subject (1921) is naturally enough completely non-Darwinian, and cannot now be regarded as anything except a period piece. One of Hering's ideas was held up to ridicule by a rival nature philosopher, Professor J. S. Haldane: the idea that a compound must exist having exactly the properties we now know to be possessed by deoxyribonucleic acid, DNA.

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